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Dengue fever prediction based on meteorological features and deep learning models

Yunyun Cheng [a,b], Rong Cheng [c], Ting Xu [c], Xiuhui Tan [c], Yanping Bai [c,*]

[a] Shanxi University of Electronic Science and Technology, Linfen, 041000, China
[b] School of Information and Communication Engineering, North University of China, Taiyuan, 030051, China
[c] School of Mathematics, North University of China, Taiyuan, 030051, China

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ABSTRACT

The dengue fever epidemic is one of the health priorities of the World Health Organization (WHO), and accurately predicting its epidemiological trends is crucial. Multi source geographic data such as temperature, humidity, and precipitation affect the occurrence and prevalence of dengue fever. Therefore, an effective hybrid model to improve the prediction performance of dengue fever is proposed considering the effects of multidimensional meteorological features. Initially, to address the issue of data scarcity, the Time-series Generative Adversarial Networks (TimeGAN) algorithm is employed to expand the dengue dataset. Second, the meteorological series are decomposed by Symplectic Geometry Mode Decomposition (SGMD), and then the sub-sequences are reconstructed into high, mid and low frequency signals by utilizing Sample Entropy (SE). Subsequently, a sliding window technique is applied to the signal band to accurately capture the key time periods affecting dengue fever. Finally, bidirectional temporal features related to the dengue sequence are extracted and fused by a bidirectional temporal convolutional network (BiTCN), and the fused features are inputted into a bidirectional long and short-term memory network (BiLSTM) introduced the attention module for prediction, and to obtain the final prediction results. Using the data of dengue fever cases in Guangdong Province, China, as an example, the experimental results show that the developed method can accurately predict the trend of dengue fever epidemic, with mean absolute error (MAE) and mean absolute percentage error (MAPE) of 192.98759 and 2.492, respectively.

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1. Background

Dengue fever is among the most rapidly spreading vector-borne diseases, posing a significant threat to human health (Chen et al., 2024). Its spread is influenced by meteorological factors, such as high temperatures and rainy seasons leading to mosquito breeding, thereby increasing the likelihood of dengue fever transmission (Xu et al., 2020). From a public health

* Corresponding author.
E-mail addresses: 1556275877@qq.com (Y. Cheng), chengro@nuc.edu.cn (R. Cheng), xuting@nuc.edu.cn (T. Xu), tanxiuhui@nuc.edu.cn (X. Tan), baiyp666@163.com (Y. Bai).

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perspective, the spread of dengue fever not only causes physical discomfort or dysfunction in patients (Lima & Laporta, 2020), but also causes panic among the population due to the prevalence of major infectious diseases, and can even stunt the economic development of a country (Zhao et al., 2020). Therefore, using new technologies to forecast the risk of dengue fever transmission could reduce its impact.

At present, the real-time public release of multi-modal epidemic data provides fundamental support for constructing data-driven predictive models, yielding significant results within the dengue fever infectious disease early warning system. Through a systematic review of the literature, dengue fever prediction methods can broadly be categorized into statistical models, dynamic models, and deep learning models.

(1) Dynamic model

Within the dengue fever early warning system, dynamic models serve as a vital tool, effectively revealing the patterns of disease transmission (Bhuju et al., 2020; Yi et al., 2021). Wang and Zhang incorporated environmental factors into the Time Series Susceptibility Infection Recovery (TSIR) model and proposed a new substantive model to analyze their impact on transmission (Wang & Zhang, 2025). Abidemi and Peter analyzed the nonlinear mathematical model of dengue thermodynamics under the conditions of symptomatic isolation and vigilance. The experimental results indicated that increasing the isolation rate for symptomatic and asymptomatic infections, as well as raising the proportion of alert individuals, can significantly reduce the disease burden of dengue fever (Abidemi & Peter, 2023). Abidemi et al. proposed a dengue fever model comprising an eight-dimensional system of autonomous ordinary differential equations. Firstly, Lyapunov functions were employed to investigate the global asymptotic dynamics of the model near the dengue equilibrium with and without the disease. Subsequently, sensitivity analysis was conducted to determine the relative importance of model parameters in disease transmission. Finally, optimal control theory was applied to extend the model into a nonlinear optimal control framework (Abidemi et al., 2022). The above research indicates that the infectious disease dynamics model has strong explanatory power in analyzing the system balance and transmission cycle of diseases. However, such models are often based on deterministic frameworks and idealized assumptions, which make it difficult to fully reflect the diverse dynamic factors that affect infectious diseases in reality. Secondly, the model structure and parameter settings are more dependent on the subjective choices of the modeler, which weakens the consistency and comparability of the model results.

(2) Statistical model

Unlike modeling infectious disease dynamics based on complex mechanisms, statistical prediction models mainly rely on historical monitoring data to establish extrapolation prediction models by identifying trends, seasonality, and periodicity in time series. A classic statistical method widely employed in dengue fever forecasting is the Autoregressive Integrated Moving Average (ARIMA) model, which predicts outcomes by applying differencing to non-stationary time series and integrating both lagged components and error terms (Hasan et al., 2024a). For instance, Patil and Pandya examined the impact of climatic factors on dengue fever incidence, employing random forests (RF), decision trees (DT), support vector regression (SVR), ARIMA, SARIMA, and Facebook Prophet to analyze and forecast dengue incidence rates across ten Indian cities (Patil et al., 2021). Naher et al. selected appropriate models from ARMA, Error-Trend-Seasonal (ETS) and Trigonometric seasonality, Box-Cox transformation, ARMA errors, Trend and Seasonal (TBATS) models to forecast dengue incidence in Bangladesh (Naher et al., 2022).

In addition to ARIMA and its derivative models, other time series models are also widely used in dengue fever modeling. Shi et al. combined case data, meteorological data, vector surveillance data and demographic data to employ the LASSO method for predicting weekly dengue incidence rates. Compared with SARIMA and descending linear regression, the LASSO model exhibited the smallest mean percentage error and the best predictive performance (Shi et al., 2016a). Gómez Gómez et al. analyzed the association between dengue cases and climatic factors in Asunción using weekly case data, then employed a generalized additive model, selecting the optimal model based on the Akaike information criterion (Gómez et al., 2022). Moreover, methods such as Spearman's correlation analysis and principal component analysis (PCA) are frequently employed to address multicollinearity or perform feature dimensionality reduction, thereby enhancing model stability and interpretability (Chen et al., 2022; Yu et al., 2025). However, traditional time series models are mostly linear and rely on the stationarity of the sequence and a fixed time-dependent structure. Therefore, their ability to capture common features such as nonlinear changes and complex mechanisms in public health emergencies is limited.

(3) Deep learning model

In recent years, with the continuous advancement of medical informatization, deep learning deep learning predictive models have been extensively applied in the field of dengue fever forecasting research (Ceia-Hasse et al., 2023; Hassan et al., 2022). Compared to traditional statistical models and early machine learning approaches, deep learning models do not rely on explicit prior assumptions regarding the relationship between input and output variables. They can automatically extract high-dimensional features through multiple layers of non-linear transformations, effectively capturing complex non-linear interactions and temporal dependencies within data. This capability demonstrates significant advantages, particularly when

processing multidimensional, high-noise medical data exhibiting intricate spatio-temporal dynamics. The early deep learning models used for dengue fever prediction were mainly based on recurrent neural networks (RNNs) and their variants (Amin et al., 2020; Jaya et al., 2025; Navarro Valencia et al., 2021). However, standard RNNs are prone to gradient vanishing or exploding issues when processing long sequences, making it difficult to effectively learn long-term dependencies. To overcome this limitation, long short-term memory (LSTM) networks and a series of extended variants have been widely introduced into this field. Kakarla et al. employed vector autoregression, generalized boosted models, SVR, and LSTM to forecast dengue fever epidemics in India. Experimental results demonstrated that the LSTM model delivered the most favourable performance, exhibiting superior predictive capabilities (Kakarla et al., 2023). Patra et al. proposed a hybrid model combining a CNN with a stacked LSTM. Results demonstrated that integrating a CNN with a single-layer stacked LSTM significantly enhances dengue prediction and can be applied to other dengue-prone regions (Patra et al., 2024).

However, within infectious disease early warning systems, even if certain predictive algorithms demonstrate superior performance in specific scenarios, no single model can maintain optimal performance across all circumstances. In recent years, the research focus has shifted from single models towards more complex and efficient hybrid architectures and novel network structures to address the unique challenges in infectious disease forecasting (Manoharan et al., 2023; Shaikh et al., 2023; Yeh et al., 2025). Anno et al. employed an encoder-decoder model utilizing U-Net to predict the spatial and temporal distribution of dengue fever outbreaks in Taiwan, China (Anno et al., 2024). Nguyen et al. used convolutional neural networks (CNN), transformer, LSTM, and attention enhanced LSTM (LSTM-ATT) models based on weather factors to predict monthly dengue fever cases in 20 provinces of Vietnam (Nguyen et al., 2022). Nanda et al. combined radial basis function networks (RBFNS) and Darts Game Optimizer (Dgo) algorithm to predict dengue fever risk (Nanda et al., 2024). In addition, some scholars have considered hybrid prediction models based on data decomposition techniques. Souza et al. proposed a novel dengue fever prediction technique. Firstly, the Laplacian pyramid was used to decompose the signal into different bands, and then diffusion mapping was used to improve the low dimensional embedding of features. Finally, SVM was used for classification (Souza et al., 2022). Anggraeni et al. proposed a hybrid model for predicting dengue fever cases by combining Empirical Mode Decomposition (EMD), Generalized Regression Neural Network (GRNN), and Particle Swarm Optimization (PSO) (Anggraeni et al., 2024).

Based on the above, we proposed a hybrid model (BiTCN-BiLSTM-Attention) by integrating bidirectional time convolutional networks (BiTCN), bidirectional long short-term memory networks (BiLSTM), and attention mechanisms. BiTCN efficiently captures long-term, multi-scale dependencies within input sequences—such as annual cycles and seasonal trends—through dilated convolutional layers. BiLSTM builds upon this foundation to model short-term dependencies and transient dynamics between sequence segments with greater precision. Finally, the attention mechanism dynamically assigns weights to distinct features and time steps, significantly enhancing the model's expressive power and interpretability.

In addition, there are still some knowledge gaps that need to be filled in the existing dengue fever prediction methods. Firstly, deep learning models have poor training performance in the case of sample size is insufficient, and appropriate methods need to be adopted to enhance the scale and quality of dengue fever data. Secondly, less studies have investigated and described the temporal dynamics between meteorological factors and dengue fever cases, namely which meteorological factors are key predictors of dengue fever at specific times. Finally, in order to improve the prediction model training speed and accuracy, some data decomposition and feature extraction methods need to be proposed to deal with meteorological feature sequences.

To address the aforementioned issues, a superposition method combining multistage feature extraction and hybrid deep learning model was developed for dengue prediction, which provided more robustness and better learning capability than traditional single model and machine learning algorithms. In the data augmentation part, a Time Series Generative Adversarial Network (TimeGAN) was used to produce high-quality meteorological and dengue samples. In the data decomposition part, symplectic geometric mode decomposition (SGMD) was applied to decompose the meteorological feature into different bands, and the sample entropy (SE) of each frequency band was calculated to reconstruct them into three signal frequency bands: high, medium, and low. In the windowing part, a sliding window was employed to decompose the signal bands to find the best time window for dengue fever prediction. In the prediction part, a hybrid deep learning prediction model was proposed to reduce the impact of the curse of dimensionality and avoid overfitting due to the added features.

2. Materials

2.1. Study area

We collected meteorological and dengue fever change data from January 2005 to April 2024 in Guangdong Province, China. The study area is located on the southeast coast, bounded by latitude 20°13′–25°31′N, and longitude 109°40′–117°30′E, as shown in Fig. 1. It is affected by a tropical monsoon and subtropical monsoon climate with hot and rainy summers and warm and dry winters, which are suitable for the survival and development of *Aedes aegypti* mosquitoes as well as the spread of dengue fever, making it a high incidence area for dengue fever.

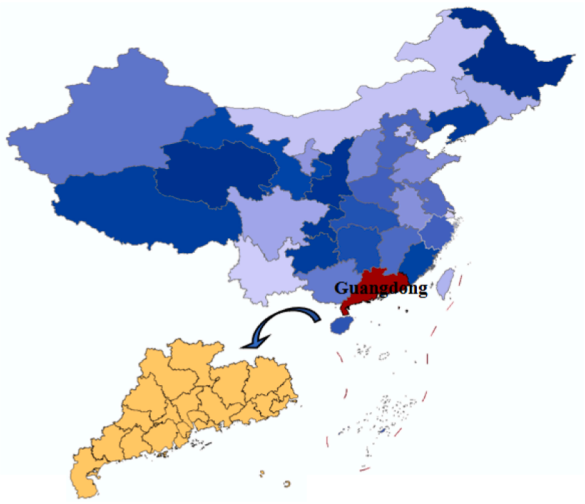


Fig. 1. Geographical location of the research area.

2.2. Data acquisition

Dengue fever case data were obtained from the China Public Health Science Data Center. Meteorological data, including daily temperature, humidity, precipitation, and sunshine duration, were sourced from the National Meteorological Science Data Center. Table 1 shows the construction process of the dataset in this study. The meteorological data for each year were constructed as a 365 × 4 feature matrix, generating a third-order feature tensor $\chi \in \mathbb{R}^{19\times365\times4}$. $\chi \in \mathbb{R}^{19\times365\times4}$ was then normalized and sliced along the annual dimension for visualization, as shown in Fig. 2, where each subplot displayed the annual meteorological evolution pattern for a single year.

Furthermore, to more effectively capture the long-term dynamic characteristics of case data, TimeGAN was employed to generate both daily frequency sequences for meteorological factors and monthly frequency sequences for dengue fever cases, maintaining a 1:4 ratio between original and synthetic data. Subsequently, the generated monthly case data were aggregated into annual case counts. Finally, the expanded dataset was partitioned into training and testing sets at an 8:2 ratio to validate the model's predictive accuracy and efficacy.

2.3. Data preprocessing

To address the problem of missing values and outliers in the dengue cases and meteorological data, we used linear interpolation of time series neighbouring observations for correction to reduce the interference of anomalous data in the model training process. Specifically, the arithmetic mean X_t was calculated based on the valid data of its adjacent time nodes (X_{t-1} and X_{t+1}) to fill in any anomalies or missing positions. The calculation formula can be expressed as:

$$X_t = \frac{X_{t-1} + X_{t+1}}{2} \tag{1}$$

in addition, the normalization method was used to map the features of each dimension to a unified scale to eliminate the risk of gradient oscillation caused by the magnitude difference between variables, while improving the convergence efficiency of the model.

Table 1
Construction steps of dataset.

Description	
Target variable	Total monthly dengue fever cases in Guangdong Province from 2005 to 2023, aggregated into annual case data
Original Features	Daily meteorological data for Guangdong Province from January 1, 2005 to December 31, 2023, including temperature, humidity, precipitation, and sunshine duration.
Feature engineering	Reorganize daily meteorological data into a time-series feature matrix by year: each year corresponds to a feature matrix of 365 × 4 meteorological variables
Frequency alignment	Align the number of cases per year from 2005 to 2023 with the complete dailymeteorological data for that year, using a third-order tensor of $\chi \in \mathbb{R}^{19\times365\times4}$ for the entire dataset

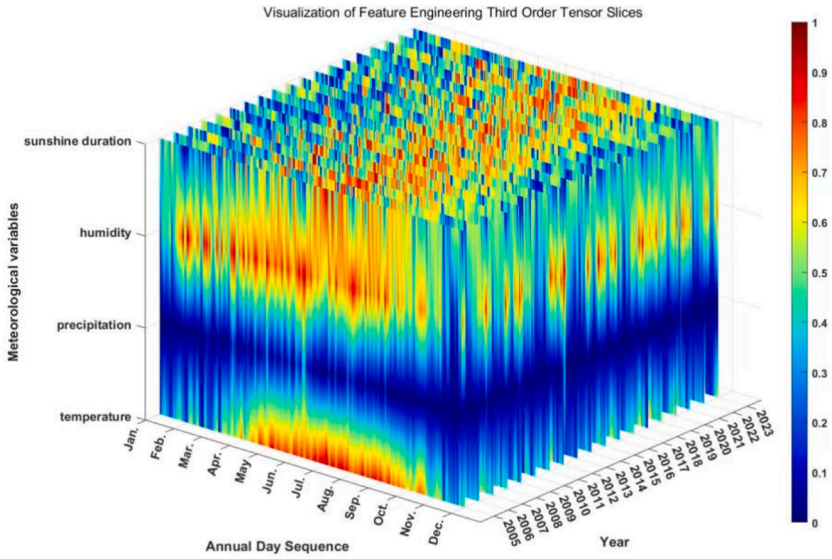


Fig. 2. Visualization of annual meteorological feature slices.

3. Methods

Fig. 3 shows the overall implementation process of our proposed model in detail. The main stages are as follows:(i) Data augmentation. Due to the limited number of years of dengue fever, TimeGAN is used to generate virtual data to expand sparse datasets and address the issue of insufficient data. (ii) Feature decomposition. The SGMD-SE decomposition and reconstruction strategy is used to reconstruct the meteorological sequence into three different signal bands to extract meteorological features at different time scales. (iii) Feature scaling (multiple time windows). Using annual meteorological data for dengue fever prediction may lose local features of seasonal fluctuations or short-term extreme weather. Therefore,

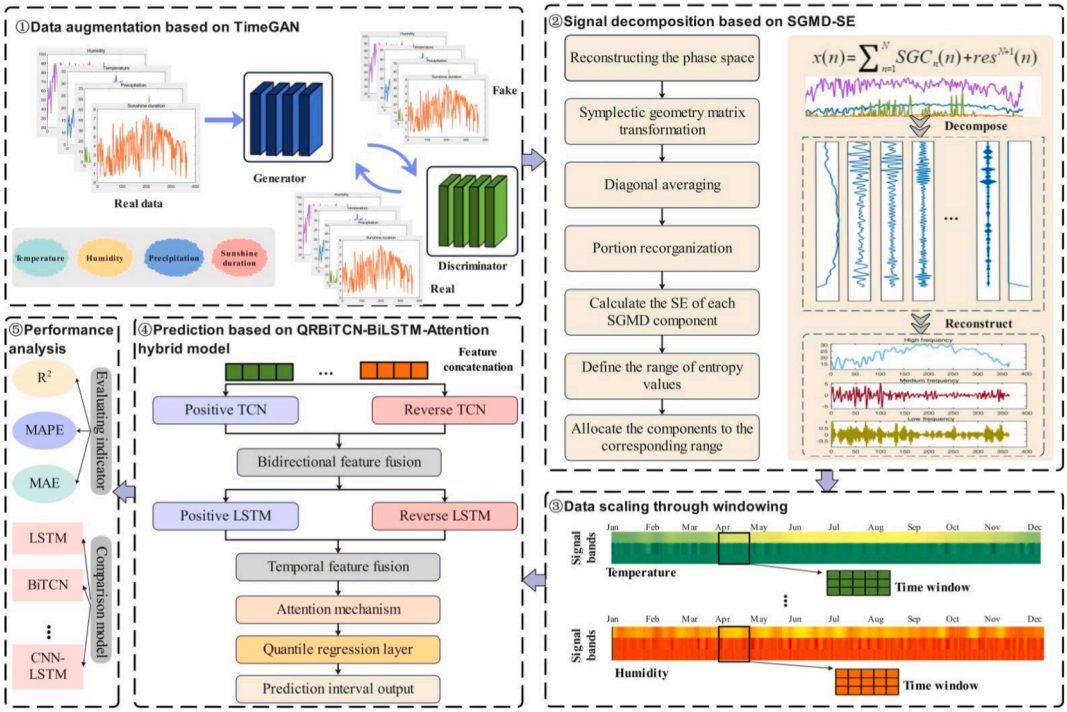


Fig. 3. The structure of the developed dengue prediction hybrid model.

we create a time window of size m to scale the feature vector and flatten and splice the selected feature into a high-dimensional vector of $r^{3\times m}$. (iv) Feature fusion and prediction. Bidirectional temporal convolutional network (BiTCN) is applied to extract hidden information related to dengue fever sequences, and the resulting information is input into bidirectional long and short-term memory network (BiLSTM) optimized by attention mechanism to obtain the final prediction result. (V) Model comparison. To verify the predictive performance of the proposed model, MAE, MAPE and R^2 are used to validate the effectiveness of each module in the hybrid model. In addition, its performance is compared with other models.

3.1. TimeGAN

TimeGAN is a variant of GAN (Yoon et al., 2019), which is built on the joint training of auto-encoder and GAN, and creatively addresses the generation and auto-regression challenges in existing data augmentation models for time series (Meng et al., 2023).

TimeGAN is composed of four key components: embedded network, recovery network, generator, and discriminator. Its core principle involves partitioning the features of time-series data into static features S that remain unchanged over time and time-series features X_T that evolve with time. It trains a latent feature distribution $\hat{p} = (S, X_T)$ to approximate as closely as possible the distribution $P = (S, X_T)$ of the original data. The specific implementation process is shown in Fig. 4 (a). The embedded network and recovery network receive raw data as input. They map the static features S and temporal features X_T of the raw data into latent codes via the embedding network, thereby generating a low-dimensional latent space. Subsequently, the recovery network remaps these latent codes back into static features and temporal features. The generator and discriminator accept noise sequences as input. The static feature z_s and temporal feature z_t of the random noise sequence are processed by the generator to produce a random latent vector in the latent space. Subsequently, the discriminator classifies this random latent vector, returning a classification probability to distinguish between real and generated data.

TimeGAN jointly trains the autoencoder and adversarial network through reconstruction loss L_R, supervised loss L_S, and unsupervised loss L_U. The training process is illustrated in Fig. 4 (b), where θ_E, θ_R, θ_G, and θ_D denote the four sets of trainable parameters. The specific formula for the loss function is as follows:

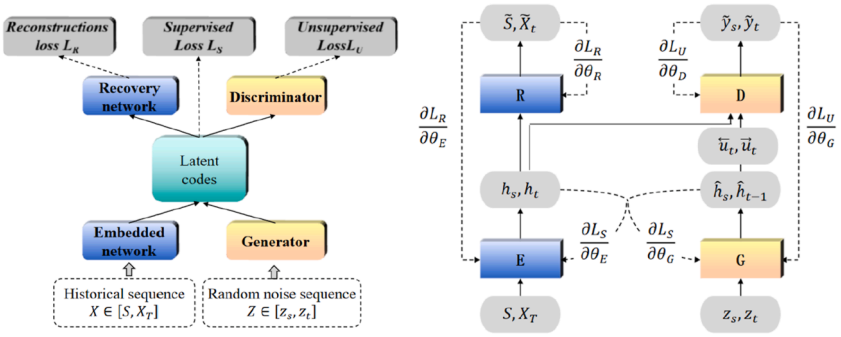
$$L_R = E_{S,X_{1:T}\sim P}\left[\|S-\tilde{S}\|_2 + \sum_t \|X_t - \tilde{X}_t\|_2\right] \tag{2}$$

where $\tilde{S}$ denotes the static feature data recovered via inverse mapping, whilst $\tilde{X}_t$ denotes the temporal feature data recovered via inverse mapping.

$$L_S = E_{S,X_{1:T}\sim P}\left[\sum_t \left\|h_t - G_\chi(h_s, h_{t-1}, z_t)\right\|_2\right] \tag{3}$$

where h_s and h_t denote the corresponding low-dimensional mappings in the latent space, while z_t represents the random noise temporal feature.

$$L_U = E_{S,X_{1:T}\sim \hat{P}}\left[ln\, y_s + \sum_t ln\, y_t\right] + E_{S,X_{1:T}\sim \hat{P}}\left[ln\,(1-\tilde{y}_s) + \sum_t ln\,(1-\tilde{y}_t)\right] \tag{4}$$



(a) Structural diagram (b) Training process

Fig. 4. TimeGAN model structure and training process.

where $\tilde{y}_s$ and $\tilde{y}_t$ denote the generator's discrimination values for static features and temporal features respectively.

3.2. Data decomposition

3.2.1. SGMD

The SGMD algorithm is an effective method for processing complex signals based on the symplectic geometric similarity transformation (SGST) (Lu et al., 2024; Peng et al., 2024). Its core idea is to obtain the eigenvalues and eigenvectors by performing the SGST on the trajectory matrix, and rebuild the individual component signal using its corresponding eigenvectors (Li et al., 2022). Compared to VMD and EMD, it has stronger resistance to mode mixing and noise. The decomposition step of SGMD is given by the following:

Step 1: Reconstructing the phase space

Let the selected time series be $x = x_1, x_2, \cdots, x_n$. By embedding Takens theory, the original time series x can be reconstructed into a phase space matrix X.

$$X = \begin{bmatrix} x_1 & \cdots & x_{1+(d-1)\tau} \\ \vdots & \ddots & \vdots \\ x_m & \cdots & x_{m+(d-1)\tau} \end{bmatrix} \tag{5}$$

where d is the embedding dimension and τ is the delay time, usually set to 1. d is defined as follows:

$$d = \begin{cases} \dfrac{n}{3}, \dfrac{f_{max}}{F_s} < 10^{-3} \\ 1.2 \times \dfrac{F_s}{f_{max}}, \dfrac{f_{max}}{F_s} \geq 10^{-3} \end{cases} \tag{6}$$

where f_{max} is the frequency of the maximum peak of the power spectral density (PSD) of x, and F_s is the sampling frequency.

Step 2: Symplectic geometric similarity transformation

The covariance matrix of X is $A = X^T X$. Construct the Hamiltonian matrix of A according to symplectic geometry theory:

$$M = \begin{bmatrix} A & 0 \\ 0 & -A^T \end{bmatrix} \tag{7}$$

Next, construct another Hamiltonian matrix $W = M^2$, and then construct a symplectic orthogonal matrix G,

$$G^T W G = \begin{bmatrix} B & R \\ 0 & B^T \end{bmatrix} \tag{8}$$

where R is the transformed sub-matrix, B is the upper triangular matrix, and its eigenvalue is $\lambda_1, \lambda_2, \cdots, \lambda_d$; The eigenvalue of A is $\sigma_i = \sqrt{\lambda_i} (i = 1, 2, \cdots, d)$, and Q_i is the eigenvector corresponding to the eigenvalue of matrix A^2. Let $Z_i = Q_i Q_i^T X$ obtain the reconstruction matrix Z:

$$Z = Z_1 + Z_2 + \cdots + Z_d \tag{9}$$

Step 3: Diagonal averaging

Convert Z_i into a one-dimensional time series $Y_i = y_1, y_2, \cdots, y_n (i = 1, 2, \cdots, d)$.

$$y_k = \begin{cases} \frac{1}{k}\sum_{p=1}^{k} z^*_{p,k-p+1}, 1 \le k \le d^* \\ \frac{1}{d^*}\sum_{p=1}^{d^*} z^*_{p,k-p+1}, d^* < k \le m^* \\ \frac{1}{n-k+1}\sum_{p=k-m^*+1}^{n-m^*+1} z^*_{p,k-p+1}, m^* < k \le n \end{cases} \tag{10}$$

where $1 \le i \le m$, $1 \le j \le d$. z_{ij} is an element in Z_i, $d^* = \min(m,d)$, $m^* = \max(m,d)$, when $m < d$, then $z^*_{ij} = z_{ij}$. Otherwise, $z^*_{ij} = z_{ji}$.

Step 4: Reorganising the portions

The normalized mean square error (NMSE) of the residual signal g^n is used to set the iteration termination condition, and the decomposition ends when the NMSE of g^n is less than a set threshold. The final decomposition result is:

$$x(n) = \sum_{n=1}^{N} SGC^h(n) + g^{N+1}(n) \tag{11}$$

where SGC is the symplectic geometric component, and N is the number of symplectic geometric components.

3.2.2. SE

SE is an improved statistical measure for analyzing time series data based on approximate entropy, which can be used to describe the irregularity and complexity of time series (Kong et al., 2023; Pang et al., 2022). A larger value of SE indicates that the time series is more similar to itself and its complexity is subsequently smaller. SE can be defined as:

$$SampleEn(m,r,L) = -\ln\left[\frac{A^m(r)}{B^m(r)}\right] \tag{12}$$

where m is the embedding dimension, r is the similarity tolerance, L is the length of the sequence, $A^m(r)$ is denotes the probability that two consecutive subsequences match to $m+1$ points under similarity tolerance r, and $B^m(r)$ denotes the probability that two consecutive subsequences match to m points under similarity tolerance r.

The SE of each SGMD component is calculated to obtain the maximum and minimum values. According to the distribution of sample entropy, the entropy value is divided into three ranges: high, medium and low. Each SGMD component is assigned to the corresponding high, medium and low entropy ranges. Finally, the signal components in each range are summed to obtain the sum of the three signal bands.

3.3. Multiple time windows

To capture when meteorological factors are key to predicting dengue, we used a sliding window technique to assess whether the most recent data or a specific time window could more accurately predict dengue epidemiological trends.

In the sliding window method, a window of suitable size was determined for searching the target data. We created a window of size *m*, defining the start of the short window as t_m and the duration as *m*. This window covered a specific time range of band data and then gradually moved this window over the signal band until the entire selected time period data was segmented. We thus obtained multiple windows, as shown in Fig. 5.

3.4. Hybrid deep learning model

The hybrid model contains three components: BiTCN, BiLSTM and attention mechanism. Fig. 6 illustrates the structure of the overall hybrid deep learning model. The BiTCN component effectively captures the front and back information of data by fusing forward and reverse temporal convolutional networks (Li & Xu, 2024; Zhang, Ye, et al., 2024). The use of different inflation factors enables the model to capture features at multiple timescales, significantly enhancing its capacity to recognize complex patterns. The data sequence processed by BiTCN is input into BiLSTM, which connects the LSTM layers in both forward and reverse directions, with one processing the input sequence in the forward direction and the other processing it in the reverse direction (He & Abisado, 2024; Lin et al., 2023; Zhang, Yang, et al., 2024). In this way, BiLSTM can explore the dependency relationship of features more comprehensively and obtain the upper and lower correlations. Attention mechanism are introduced into BiLSTM, which adaptively focuses on the features that are most important for

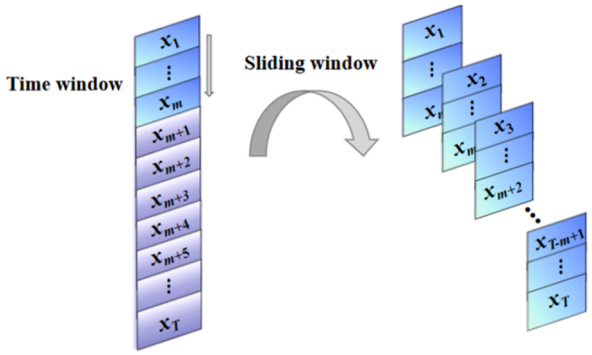


Fig. 5. Sliding window schematic.

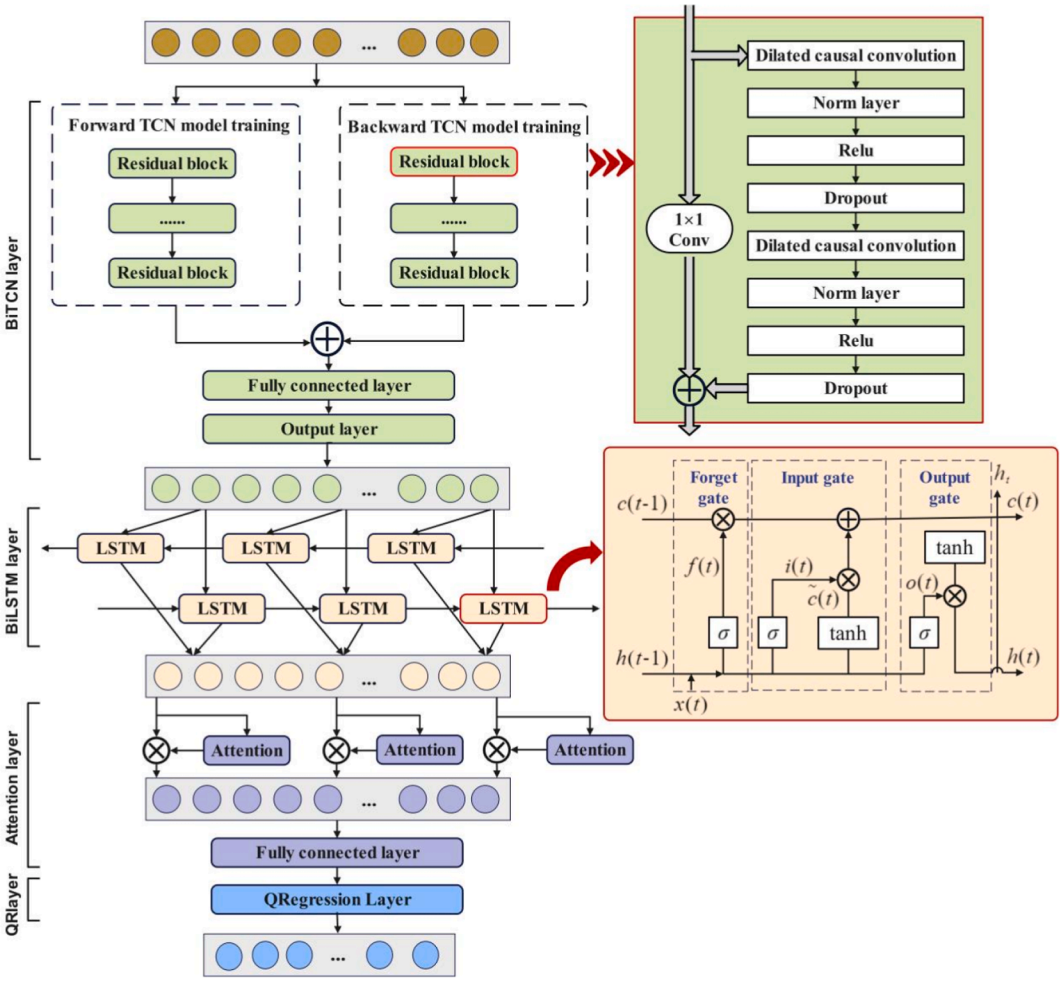


Fig. 6. The structure of BiTCN-BiLSTM-Attention.

current task (Li et al., 2024; Oyewola et al., 2024). This not only improves the accuracy of the model, but also increases the model's interpretability of its decision-making process.

4. Experimental results and analysis

4.1. Parameter setting

In this study, the meteorological feature data were decomposed into several components of different frequencies using SGMD, denoted as SGC_1, SGC_2, … SGC_{N-1}, RES, as shown in Fig. 7(a). We then reconstructed each component into three signal frequency bands using SE, as shown in Fig. 7(b). Table 2 presents the prediction errors of SGMD with different N values. The experimental results indicated that the optimal range for N was 6–11. If N is too small, the model may fail to capture key patterns; if too large, overfitting to noise can occur. For example, the MAE reached its minimum value of 192.987 at $N = 10$. Table 3 summarizes the parameter settings of the hybrid deep learning model, including the Adam optimizer and a maximum iteration count of 100.

4.2. Evaluation metrics

To accurately quantify the predictive performance of different models, three evaluation metrics of mean absolute error (MAE), mean absolute percentage error (MAPE) and fitting degree (R^2) were adopted in this study. The formulae are as follows:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{13}$$

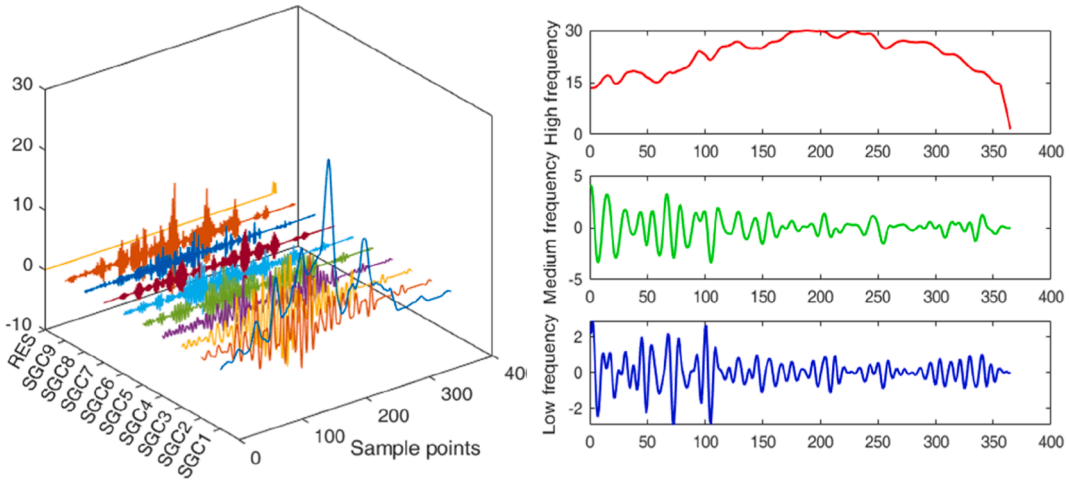
$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100\% \tag{14}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y}_i)^2} \tag{15}$$

where n is the number of samples, y_i is the actual number of dengue fever cases, $\hat{y}_i$ is the predicted number of dengue fever cases, and $\bar{y}_i$ is the average value.

4.3. Data augmentation effectiveness verification

The TimeGAN framework was employed to generate meteorological factors and dengue fever cases. An expanded dataset was generated using a 1:4 ratio of original data to synthetic data. As meteorological data constitutes a relatively exogenous variable, its variations are primarily governed by natural laws. Consequently, it could be modelled and generated



(a) SGMD decomposition results (b) SE refactoring results

Fig. 7. Decomposition and reconstruction results based on SGMD-SE.

Table 2
Prediction errors of SGMD on dataset with different N values.

N	MAE	MAPE	R^2
6	491.709	7.949	0.997
7	540.675	8.837	0.994
8	441.067	3.699	0.997
9	399.005	4.482	0.997
10	**192.987**	**2.492**	**0.999**
11	367.582	3.930	0.997

Table 3
Other parameter settings for the model.

Model	Parameters	Value
BiTCN	Num Filters	64
	Filter Size	5
	Dropout Factor	0.1
	Num Blocks	1
BiLSTM	Max Epochs	100
	Gradient Threshold	1
	Initial Learn Rate	0.001
	L2 Regularization	0.0001
	Execution Environment	cpu

independently of dengue fever case data. Its input dimensions were $[N_1, T_1, D_1]$, where N_1 denoted the sample size of the original meteorological sequence, T_1 represented the time step, set to 365, and D_1 signified the feature dimension, set to 4. Additionally, to capture long-term characteristics of case sequences, modeling was performed using the monthly frequency distribution of dengue fever cases. The input dimensions were $[N_2, T_2, D_2]$, where N_1 denoted the number of original monthly case sequences, T_2 represented the time step (set to 12 to correspond with the annual cycle), and D_2 signified the feature dimension comprising dengue case counts plus four meteorological features, totalling five dimensions. The generated monthly case data was subsequently aggregated into annual case data to align with the annual scale of meteorological data. Finally, the original data was concatenated with the generated data to form an expanded dataset, which was then split into training and test sets in an 8:2 ratio for subsequent model training and validation. Fig. 8 illustrates a comparison between selected synthetic data points and genuine data samples. The results demonstrate that the generated data aligns with the overall trends of the original data while maintaining reasonable local variability. This indicates that TimeGAN has successfully learned the distributional characteristics of the original data rather than merely replicating the original sequences.

In order to scientifically evaluate the intrinsic relationship between synthetic data and original data, PCA and t-SNE dimensionality reduction techniques were used for quantitative evaluation. As shown in Fig. 9 (a), although the synthetic data did not completely cover the entire distribution of the original data, the two exhibited substantial overlap along the first principal component (PC1), which accounts for 69.3 % of the cumulative variance. The second principal component (PC2) explains 37.2 % of the variance, significantly less than that of PC1. These results demonstrate that the synthetic data successfully preserves the major variation patterns of the original sequences. Fig. 9 (b) displays the visualization results from t-SNE, revealing extensive overlap between the two datasets in the low-dimensional space, with highly similar contour distributions. Notably, the synthetic and original data are closely intertwined across multiple high-density regions, such as the distribution centres at the edges and the transition zones. The experimental results demonstrate that the generated data effectively captures the intrinsic distributional characteristics of the original dataset.

4.4. Comparisons of windowing effects

In this study, we did not use annual meteorological data as feature vectors. Instead, the optimal start date and duration were determined through a time window. Six different sized windows were set: 20, 30, 50, 75, 100, 150. Fig. 10 and Table 4 show the predictive performance of the developed model under different time windows. The experimental results showed that all time windows achieved better accuracy than the entire annual cycle window, and the algorithm exhibited superior prediction performance when selecting June, July, and August meteorological data for prediction.

4.5. Comparison with single model

Table 5 shows the performance of seven models used for dengue fever prediction. To assess the BiTCN-BiLSTM-Attention model, we compared it with single models (TCN, LSTM, BiLSTM, BiTCN) and hybrid LSTM-TCN models (TCN-LSTM and BiTCN-BiLSTM). Compared with TCN-LSTM, the BiTCN-BiLSTM model reduced MAE and MAPE by 18.71 % and 6.18 %, respectively, and increased R^2 by 0.2 %. These improvements indicate that the bidirectional structure extracts depth

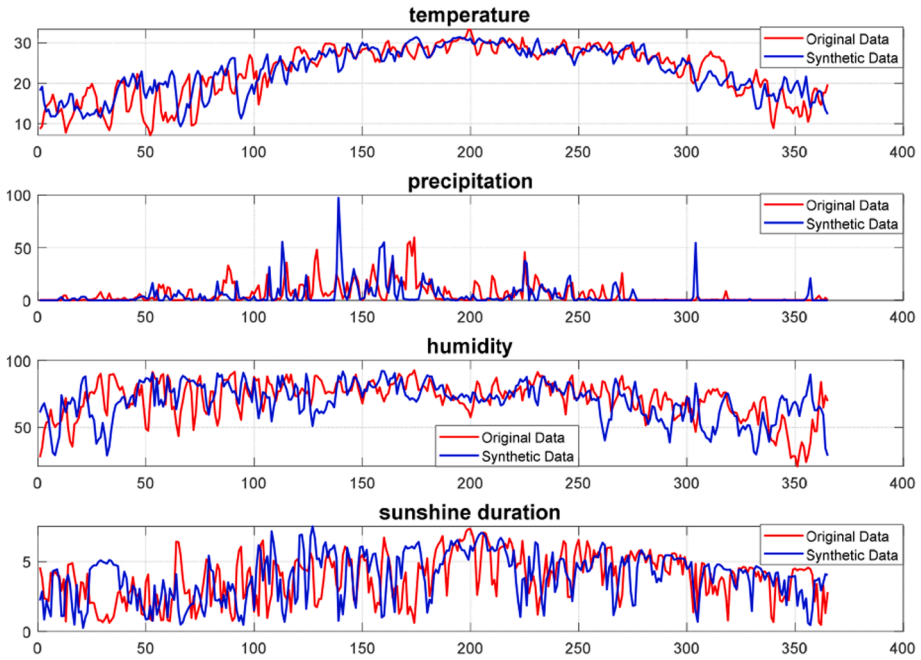


Fig. 8. Comparison of partial sample curves of generated data and real data for various meteorological factors.

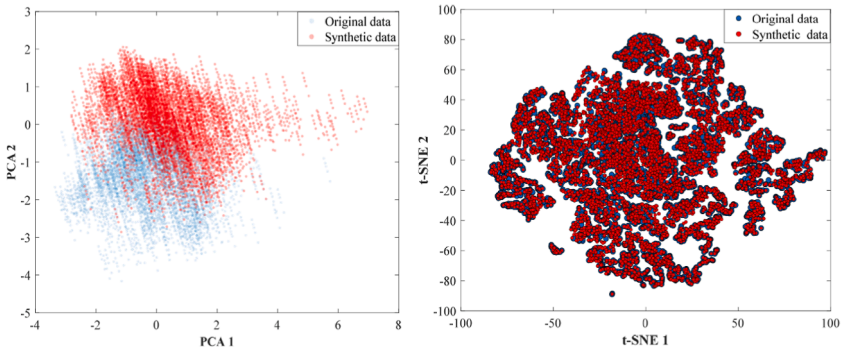


Fig. 9. PCA and t-SNE analysis results.

information more effectively than the unidirectional structure. In addition, compared with BiTCN-BiLSTM, the MAE and MAPE of BiTCN-BiLSTM-Attention decreased by 24.06 % and 25.20 %, respectively, and R^2 increased by 0.08 %. This demonstrates that the attention mechanism can make the model more focused on the features that have the greatest impact on the results through dynamic weighting, thereby improving the accuracy and robustness of predictions.

4.6. Comparisons of decomposition effects

Fig. 11 shows the prediction results of three prediction models before and after SGMD-SE decomposition. The results indicated that the predicted number of dengue fever cases using each method was consistent with the actual number of cases. Table 6 presents the error results of each comparison model. After SGMD-SE decomposition, the MAE of the three prediction models decreased by 8.01 %, 7.17 %, and 24.12 %, respectively. MAPE decreased by 22.04 %, 7.52 %, and 6.10 %, respectively. R^2 increased by 0.04 %, 0.06 %, and 0.07 % respectively. The results showed that compared with the original model, SGMD-SE decomposed meteorological sequences into multiple smooth components, removed redundancy and noise, extracted key information related to dengue fever cases, and provided more concise and effective input features for the model.

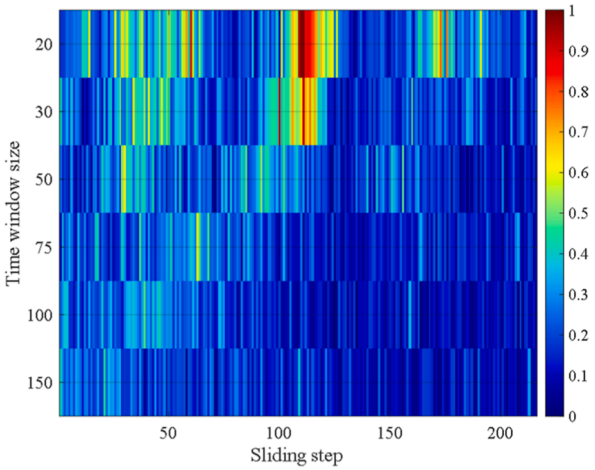


Fig. 10. Normalized MAPE matrix heatmap.

4.7. Comparison with other models

To further validate the effectiveness of the developed dengue prediction method (SGMD-SE-Window-BiTCN-BiLSTM-Attention), the proposed model was compared with other methods (Backpropagation network (BP), RF) and hybrid methods (PSO-BP, CNN-LSTM, VMD-BP).

Table 7 shows the prediction errors of the proposed model and other models. As shown in Table 7, the developed method had the lowest MAE and MAPE values, and the highest R^2 values of 192.987, 2.492, and 0.999, respectively, which can accurately predict dengue fever cases. Taking MAPE as an example, the prediction error of dengue fever based on BP was the largest, with a MAPE of 16.586. Compared with VMD-BP, The MAPE and MAE of the developed method were reduced by 52.126 % and 61.004 %, respectively, and the R^2 was increased by 0.245 %. In addition, the prediction error of dengue fever based on CNN-LSTM was still higher than that of the proposed hybrid method. This indicated that although the CNN-LSTM model can effectively extract spatial features and capture temporal dependencies, relying solely on the combination of these two models may not fully optimize prediction results in the face of complex spatiotemporal data and high noise.

By quantifying the relationship between predicted and actual values through R^2, the fit of the developed dengue fever prediction method can be effectively identified. Fig. 12 presents the regression scatter plots and confidence intervals of actual and predicted values under different prediction models. The narrower the width of the confidence interval, the better and more stable the prediction accuracy of the model. As shown in Fig. 12, the proposed method achieves the narrowest confidence interval width, demonstrating that it provides the highest prediction accuracy among all models.

5. Discussion

Meteorological factors are effective predictors of dengue fever (Francisco et al., 2024), such as temperature (Lana et al., 2018), relative humidity (Karim et al., 2012), precipitation (Choi et al., 2016), and sunshine duration (Shi et al., 2016b) have been widely used in dengue fever prediction systems. This study achieved knowledge transfer from “environmental monitoring” to “epidemic warning” by integrating multi-scale meteorological features and deep learning models, providing a new paradigm for predicting insect borne infectious diseases.

5.1. Impact of meteorological factors at different scales on dengue fever prediction

Traditional dengue forecasting models such as Hasan et al. relied only on historical case data for dengue forecasting using ARIMA model (Hasan et al., 2024b). Patil and Pandya directly used raw meteorological data as inputs and analyzed the relationship between temperature and cases through multiple regression models, ignoring the multiscale nature of meteorological variables (Patil & Pandya, 2021). We adopted the SGMD-SE decomposition reconstruction strategy to model from "single time series" to "multimodal meteorological coupling", and revealed the nonlinear driving mechanism of meteorological signals in different frequency bands on the epidemic. In addition, compared with the traditional VMD-BP decomposition method, the MAE of the proposed model was reduced by 52 %, which can more accurately capture the impact of different frequency bands of meteorological signals on dengue fever epidemic. For example, the high-frequency component can reflect the impact of short-term temperature fluctuations on mosquito vector activity, and the low-frequency component corresponds to seasonal climate change.

Table 4
Prediction error of the developed model over different time windows.

Start date	Window size(day)	MAE	MAPE	R^2
August 7	20	230.342	2.280	0.999
March 20	30	562.911	6.775	0.992
July 3	50	294.921	1.388	0.998
August 4	75	334.678	2.328	0.997
May 14	100	230.355	1.350	0.998
June 18	**150**	**192.987**	**2.492**	**0.9995**
January 1	365	784.612	17.879	0.991

Table 5
Prediction error of each single model.

Model	MAE	MAPE	R^2
TCN	805.641	14.266	0.988
LSTM	571.628	4.329	0.994
BITCN	443.672	5.066	0.996
BILSTM	438.859	3.849	0.995
TCN-LSTM	356.728	3.611	0.997
BiTCN-BiLSTM	270.914	2.701	0.998
BiTCN-BiLSTM-Attention	**254.331**	**2.653**	**0.998**

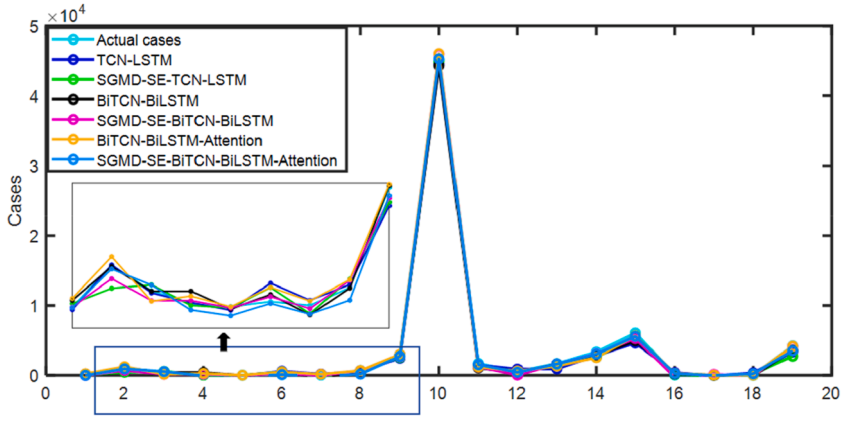


Fig. 11. Prediction results of different models after SGMD-SE decomposition.

Table 6
The prediction errors of different models after SGMD-SE decomposition.

Model	MAE	MAPE	R^2
TCN-LSTM	356.728	3.611	0.997
SGMD-SE-TCN-LSTM	328.161	2.815	0.997
BiTCN-BiLSTM	270.914	2.701	0.998
SGMD-SE-BiTCN-BiLSTM	251.477	2.498	0.998
BiTCN-BiLSTM-Attention	254.331	2.6538	0.998
Proposed model	**192.987**	**2.492**	**0.9995**

Table 7
Prediction errors of other models.

Model	MAE	MAPE	R^2
BP	489.871	16.586	0.994
RF	549.764	9.220	0.995
PSO-BP	479.439	6.822	0.996
CNN-LSTM	381.746	5.361	0.996
VMD-BP	403.122	6.390	0.997
Proposed model	**192.987**	**2.492**	**0.9995**

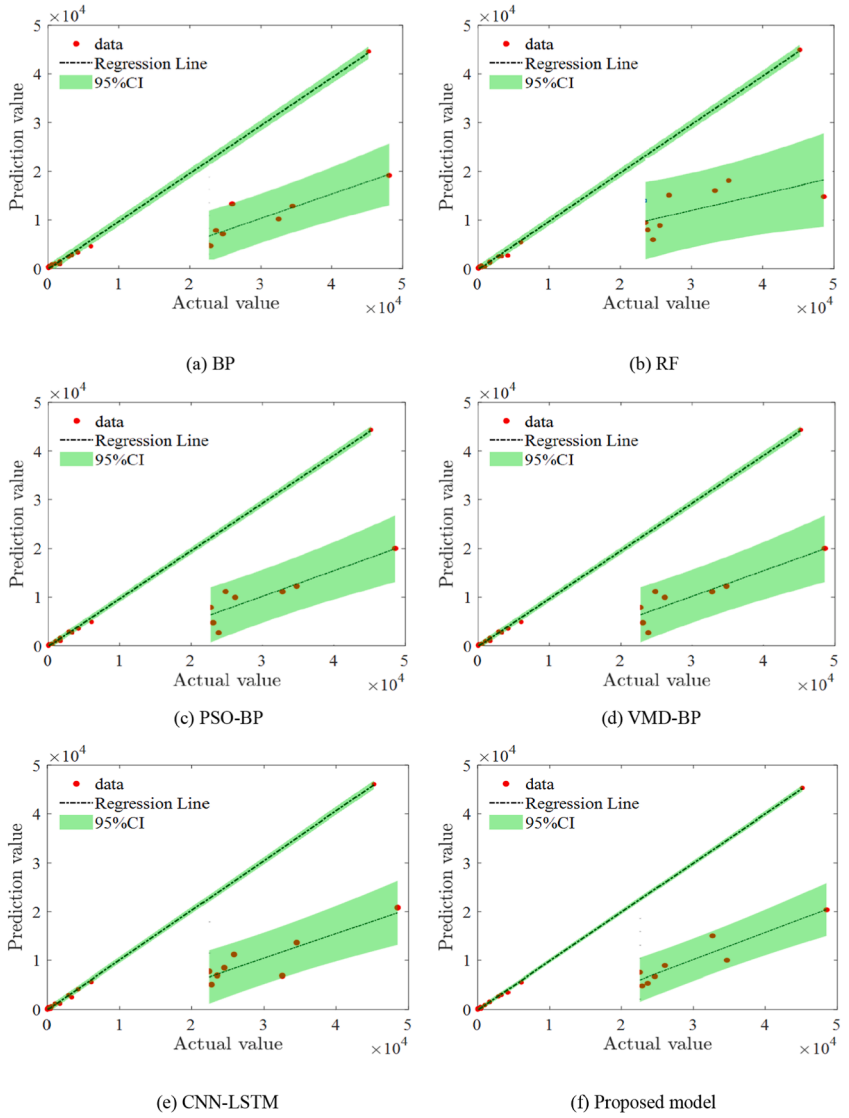


Fig. 12. Regression scatter plot of actual and predicted values using different prediction methods.

5.2. Enhancement of predicting model performance by meteorological feature engineering

Possible shortening of mosquito vector life cycles in the context of a warming climate (Ryan et al., 2019). The sliding window framework allows dynamic adjustment of key time periods through periodic retraining, avoiding performance degradation of fixed models due to environmental changes. In this study, the MAE and MAPE of the prediction model were lower with the addition of windows compared to the prediction of dengue fever using the entire annual cycle. And the meteorological factors in the months of June, July and August had the greatest impact on dengue fever. In addition, the hybrid BiTCN-BiLSTM-Attention framework achieved dynamic correlation modeling of long-term climate evolution patterns with short-term meteorological fluctuations. Compared with other prediction models such as LSTM (Majeed et al., 2023) and RF (Zhao et al., 2020), it can more comprehensively extract effective information between dengue fever cases and meteorological sequences, and has better prediction performance and stability.

5.3. The application value of dengue fever prediction model

The predictive model developed in this study enables public health agencies to forecast dengue incidence trends, facilitating data-driven decisions for vaccine distribution and dynamic allocation of vector control resources. Specifically, it transforms meteorologically sensitive "critical transmission periods" into actionable prevention windows, allowing pre-

emptive implementation of targeted interventions to mitigate outbreak risks. Although the experiment is based on data from Guangdong Province, the method framework can be extended to other tropical/subtropical regions (such as Southeast Asian countries). Inter-regional differences in mosquito vector species and population immune background may affect the lag period of meteorological factors, and parameter adjustments should be made in conjunction with local data during migration.

5.4. Limitations and future work

The current model does not include socio-economic factors (such as population mobility, urban greening rate) and real-time mosquito vector monitoring data. In the future, the prediction accuracy can be further improved through multi-source data fusion. The sliding window needs to traverse multiple time periods, and the training time increases by 2.5 times compared with the fixed model, and an evolutionary algorithm can be introduced to optimize the window search efficiency in the future. Although the proposed model performs well in Guangdong Province, its applicability in other regions needs further validation.

6. Conclusion

This study confirms the effectiveness of fusion data augmentation, multi-scale feature engineering and hybrid deep learning in dengue fever prediction, providing an innovative paradigm of decomposition-reconstruction-multi-window fusion in the field of time-series prediction. However, the ultimate goal of dengue fever prevention and control is to improve human health through predictive results, which requires model developers to collaborate with public health practitioners to truly realize the public health value of predictive science. Future work will integrate real-time meteorological forecasts and mosquito vector monitoring data to build a "time adaptive early warning system", and explore the extended application of this framework in other climate sensitive infectious diseases such as malaria and Zika.

CRediT authorship contribution statement

Yunyun Cheng: Writing – original draft, Software, Methodology. **Rong Cheng:** Validation, Investigation, Funding acquisition. **Ting Xu:** Software, Methodology. **Xiuhui Tan:** Formal analysis, Data curation. **Yanping Bai:** Supervision, Methodology, Funding acquisition.

Data availability statement

All data used to generate results was collected by the author from daily announcements issued by government departments, health committees, and meteorological stations. The data in this study can be obtained from Supplementary material.

Ethical approval

Not applicable.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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